# AMIRHOSSEIN GHAFFARI

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## EDUCATION

**Doctor of Philosophy (PhD) in Computer Science and Engineering** *May 2023 - Present*
University of Oulu, Oulu, Finland
GPA: 4.5 / 5

**Master of Science (M.Sc.) in Electrical Engineering (Digital Electronic Systems)** *2017 - 2019*
Sharif University of Technology, Tehran, Iran
GPA: 18.20 / 20

**Bachelor of Science (B.Sc.) in Electrical Engineering** *2012 - 2017*
Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran
GPA: 16.24 / 20

**High School Diploma in Mathematics and Physics** *2008 - 2012*
Roshd1 Educational Complex, Tehran, Iran
GPA: 19.87 / 20

## RESEARCH INTERESTS

Urban Computing
Machine Learning
Graph Neural Networks
Federated Learning
Generative AI

## RESEARCH AND PROFESSIONAL EXPERIENCE

**Visiting Researcher** *September 2026 - November 2026*
*Technical University of Denmark (DTU)* *Kongens Lyngby, Denmark*

· Research visit to the Machine Learning for Smart Mobility (MLSM) group, hosted by Prof. Francisco Camara Pereira.

**Doctoral Researcher** *May 2023 - Present*
*University of Oulu* *Oulu, Finland*

· Conducting doctoral research in urban computing, machine learning, graph neural networks, multi-source urban data, and data-driven analysis of smart city systems.

· Designing and evaluating spatio-temporal graph neural network architectures for large scale urban data, implemented in Python with PyTorch and PyTorch Geometric on GPU computing clusters.

· Author and co-author of peer reviewed publications in international conferences and journals; the full list appears in the Publications section.

**Computer Vision and Deep Learning Researcher** *December 2019 - April 2023*
*Fard Iran Co.* *Tehran, Iran*

· Designed and implemented an algorithm for an Automatic Number Plate Recognition system.

· Designed and implemented motion detection and human detection applications for surveillance systems.

· Designed and implemented an algorithm to automatically optimize camera parameters, including gain and shutter speed, ensuring the capture of high quality car license plate images throughout the day and night.

· Designed and implemented a signal processing tool using neural networks to measure vehicle weights in a Weigh In Motion (WIM) system.

· Designed and implemented a driver assistance tool to detect abnormal driving using deep learning and GPS sensor data for comprehensive driving analysis.

· Contributed to the large scale successful deployment of more than 70 Weigh In Motion (WIM) sites across Iran; participated in on-site installation, system debugging, performance optimization, and technical support to ensure operational reliability.

**Deep Learning Researcher** *February 2020 - October 2022*
*Institute of Water and Energy, Sharif University of Technology* *Tehran, Iran*

· Designed and implemented algorithms using deep learning and machine learning methods for an air pollution prediction system.

**Engineering Intern** *July 2016 - September 2016*
*Mahan Factory* *Isfahan, Iran*

· Completed an industrial summer internship as part of the B.Sc. program in Electrical Engineering.

· Worked with the engineering team on electrical and electronic systems used in the production line, including measurement, testing, and documentation tasks.

## THESIS PROJECTS

**M.Sc. Thesis** *2017 - 2019*
*360-degree Video Transmission over Network using Coded Caching*

Supervisor: Assoc. Prof. Mohammad Sharifkhani

**B.Sc. Thesis** *2016 - 2017*
*Design and Fabrication of an Active Noise Cancellation System for Environmental Disturbing Sounds*

Supervisor: Prof. Hassan Kaatuzian

## TEACHING EXPERIENCE

**Teaching Assistant** *Spring 2024, Spring 2025, Spring 2026*
*Big Data Processing and Applications, M.Sc. Course* *University of Oulu*

· Faculty of Information Technology and Electrical Engineering (ITEE)

· Instructor: Adj. Prof. Ekaterina Gilman

**Teaching Assistant** *Spring 2025*
*Network Analysis and Graph Machine Learning, M.Sc. Course* *University of Oulu*

· Faculty of Information Technology and Electrical Engineering (ITEE)

· Instructor: Prof. Joydeep Chandra

**Teaching Assistant** *Spring 2019*
*Computer Vision, M.Sc. Course* *Sharif University of Technology*

· Electrical Engineering Department

· Instructor: Assoc. Prof. Hoda Mohammadzadeh

**Teaching Assistant** *Fall 2018*
*Computer Architecture, B.Sc. Course* *Sharif University of Technology*

· International Campus

· Instructor: Asst. Prof. Hamed Farbeh

**Teaching Assistant** *Fall 2018*
*Advanced Computer Architecture, M.Sc. Course* *Sharif University of Technology*

· Electrical Engineering Department

· Instructor: Dr. Mohammadreza Movaheddin

**Mathematics and Physics Teacher** *2012 - 2015*
*Roshd1 Educational Complex* *Tehran, Iran*

· Taught mathematics and physics at high school level.

## PUBLICATIONS

- **A. Ghaffari**, A. Goodarzi, H. Nguyen, S. Hosio, L. Lovén, and E. Gilman, "Redditpersona: A modular framework for community-conditioned llm adaptation from reddit," in *Proceedings of the 35th ACM International Conference on Information and Knowledge Management (CIKM)*, Rome, Italy: Association for Computing Machinery, Nov. 2026. DOI: 10.1145/3799682.3840182.
- **A. Ghaffari**, H. Nguyen, L. Lovén, and E. Gilman, "Stram: Spatio-temporal road-aware mapping for graph neural network prediction," *Neurocomputing*, vol. 696, p. 134 161, 2026.
- **A. Ghaffari**, S. Pirttikangas, and E. Gilman, *Spectral adaptive graph transformer for efficient and scalable traffic forecasting*, 2026. arXiv: 2610.25140.
- **A. Ghaffari**, S. Sheikhi, and E. Gilman, "Graph-conditioned mixture of graph neural network experts for traffic forecasting," in *2026 27th IEEE International Conference on Mobile Data Management (MDM)*, 2026, pp. 1–9. DOI: 10.1109/MDM71479.2026.00037.
- D. Goncharenko, **A. Ghaffari**, and E. Gilman, "Open-data traffic noise prediction across cities and traffic stations," in *2026 IEEE 42nd International Conference on Data Engineering Workshops (ICDEW)*, IEEE, 2026, pp. 231–239.
- H. Nguyen, M. Bettinelli, **A. Ghaffari**, A. Benoit, H.-T. Nguyen, S. Pirttikangas, and L. Lovén, "From data heterogeneity to convergence: A data-centric review of federated learning," *Neurocomputing*, p. 135 005, 2026.
- H. Nguyen, **A. Ghaffari**, J. Chandra, E. Gilman, and L. Lovén, "Centrality matters: Lightweight graph-centric aggregation for moderately heterogeneous federated learning," *IEEE Transactions on Artificial Intelligence (TAI)*, 2026.
- H. Nguyen, **A. Ghaffari**, and L. Lovén, "When we talk morse: Lightweight parameter encoding for secure federated learning," in *Availability, Reliability and Security. ARES 2026 International Workshops*, Cham: Springer Nature Switzerland, 2026, pp. 447–465.
- H. Ahmed, H. Nguyen, **A. Ghaffari**, E. Gilman, and L. Lovén, "Past to plan: Llm-powered personalized travel via mobility patterns," in *13th IEEE International Conference on Big Data (BigData)*, IEEE, 2025, pp. 7652–7661.
- **A. Ghaffari**, H. Nguyen, L. Lovén, and E. Gilman, "Stm-graph: A python framework for spatio-temporal mapping and graph neural network predictions," in *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25)*, Seoul, Republic of Korea: Association for Computing Machinery, Nov. 2025, pp. 6377–6381. DOI: 10.1145/3746252.3761645.
- **A. Ghaffari**, S. Pirttikangas, and E. Gilman, "Understanding well-being in urban context: A survey," *IEEE Access*, vol. 13, pp. 11 136–11 158, 2025. DOI: 10.1109/ACCESS.2024.3506935.
- S. Sheikhi, **A. Ghaffari**, A. Amiri, and L. Lovén, "Federated variational autoencoders for unsupervised anomaly detection in distributed 5g networks," in *33rd International Conference on Software, Telecommunications and Computer Networks (SoftCOM)*, IEEE, 2025, pp. 1–6.
- S. Sheikhi [*], A. Amiri [*], and **A. Ghaffari** [*], "Balancing Latency and Energy: A Hybrid Geometric Method for Leader Drone Positioning in IoT-Enabled Smart Cities," in *22nd International Conference on Networking, Sensing and Control (ICNSC)*, [*]Equal contribution, IEEE, 2025, pp. 364–369.
- **A. Ghaffari**, "Urban well-being: Leveraging multi source data for informed decision-making," in *22nd IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops)*, IEEE, 2024, pp. 358–359.
- **A. Ghaffari**, H. Nguyen, A. Saleh, L. Lovén, and E. Gilman, "Traffic accident prediction and warning system: Integration use case," in *Fourth Workshop on Knowledge-infused Learning (KIL), in conjunction with the 30th ACM conference for Knowledge Distillation and Data Mining (KDD)*, 2024.

## CONFERENCES, WORKSHOPS, AND SUMMER SCHOOLS

- **ACM CIKM 2026**, Rome, Italy. First author and presenter of a paper, Chair of "the 2nd International Workshop on Graph-Augmented LLMs (GaLM): Bridging Language and Structured Knowledge" workshop at CIKM. *Fall 2026*
- **ARES 2026**, Linköping, Sweden. Co-author and oral presenter of a paper. *Summer 2026*
- **IEEE MDM 2026**, Athens, Greece. First author and oral presenter of a paper. *Summer 2026*
- **ACM CIKM 2025**, Seoul, South Korea. Participated as a student volunteer and first author presenter

of a paper. *Fall 2025*

- **IEEE ICNSC 2025**, Oulu, Finland. Participated as an author of a paper. *Fall 2025*
- **NVIDIA Workshop**, University of Oulu, Finland. Participated in the NVIDIA workshop on Fundamentals of Accelerated Computing with CUDA Python. *Fall 2024*
- **ACM IoT 2024**, Oulu, Finland. Participated as a student volunteer helping the organizing committee. *Fall 2024*
- **ACM KDD 2024**, Barcelona, Spain. Full participation with a poster presentation. *Fall 2024*
- **12$^{th}$ UBISS Summer School**, University of Oulu, Finland. Full participation and worked as a liaison student helping organizers in the "Care-full! More-than-human Urban Futures" workshop. *Summer 2024*
- **6G Symposium**, Levi, Finland. Full participation. *Spring 2024*
- **IEEE PerCom 2024**, Biarritz, France. Full participation with a poster presentation. *Spring 2024*
- **11$^{th}$ UBISS Summer School**, University of Oulu, Finland. Full participation in the "Smart Internet of Things (IoT) Wireless Connectivity in 5G and Beyond" workshop. *Summer 2023*
- **IEC2-2019 Sharif**, Sharif University of Technology, Tehran, Iran. Attended KNIME, Tableau, and Pandas workshops. *Summer 2019*
- **Deep Learning Summer School**, University of Tehran, Tehran, Iran. Full participation. *Summer 2018*
- **TensorFlow Workshop**, Sharif University of Technology, Tehran, Iran. Full participation. *Summer 2018*

## AWARDS AND HONORS

- ACM CIKM Student Travel Grant for the 34th ACM International Conference on Information and Knowledge Management (CIKM), Seoul, Republic of Korea, 2025.
- Tauno Tönning Foundation personal research grant, 2024.
- IEEE Student Travel Grant for the IEEE International Conference on Pervasive Computing and Communications (PerCom), Biarritz, France, 2024.
- Four year fully funded doctoral researcher position through competitive selection at the University of Oulu, 2023.
- Iran's National Elites Foundation Award, 2019.
- National graduate and undergraduate full scholarships.
- Ranked within the top 0.5 percent among more than 31,000 participants in the Iranian Electrical Engineering graduate university entrance exam.
- Ranked within the top 0.5 percent among more than 220,000 participants in the Iranian Mathematics and Physics university entrance exam.

## CERTIFICATIONS AND COURSES

- **Professional Certifications and Courses**, Neo4j GraphAcademy
  Profile: graphacademy.neo4j.com
- **Knowledge Graphs: Foundations and Applications**, openHPI *Issued April 2026*
  Certificate: openHPI
- **Foundation: Introduction to LangGraph (Python)**, LangChain Academy *Issued December 2025*
  Certificate: academy.langchain.com/certificates/os5jif5euk
- **AI Agents Intensive Course**, Google *Issued November 2025*
- **Generative AI Intensive Course**, Google *Issued September 2025*
- **Fundamentals of Accelerated Computing with CUDA Python**, NVIDIA *Issued October 2024*
  Certificate: learn.nvidia.com/certificates
- **Machine Learning Specialization**, Coursera *Issued October 2022*
  Certificate: coursera.org/account/accomplishments

- **Professional Project-Oriented Course in Backend Development with Django**, Quera *Issued December 2021*
  Certificate: quera.org/certificate/dW9TlTmL
- **Deep Learning Summer School**, University of Tehran *Issued August 2018*
  Certificate: available on request
- **Deep Learning with TensorFlow**, Sharif University of Technology *Issued July 2018*
  Credential ID: AC-97-EE-10001

## TECHNICAL SKILLS

- **Deep Learning and Graph Neural Networks:** PyTorch, PyTorch Geometric, Hugging Face, TensorFlow.
- **Agentic AI and Large Language Models:** LangGraph, LangChain, Google ADK, OpenAI and other LLM provider APIs, RAG and GraphRAG pipelines.
- **Big Data and Distributed Systems:** Apache Spark, PySpark.
- **Databases and Knowledge Graphs:** PostgreSQL, Neo4j, Cypher, SPARQL, RDF.
- **MLOps and Cloud Native:** Docker, Kubernetes, Prometheus, Grafana, Weights and Biases, Git.
- **Programming Languages:** Python, C, C++, MATLAB.

## LANGUAGES

- **Persian:** Native
- **English:** Full professional proficiency
- **Finnish:** Beginner
- **Swedish:** Beginner

## REFERENCES

- **Ekaterina Gilman**
  Adjunct Professor (Docent), Faculty of Information Technology and Electrical Engineering (ITEE), University of Oulu, Oulu, Finland
  Email: ekaterina.gilman@oulu.fi
  Web Page: oulu.fi/en/researchers/ekaterina-gilman
- **Susanna Pirttikangas**
  Adjunct Professor (Docent), Faculty of Information Technology and Electrical Engineering (ITEE), University of Oulu, Oulu, Finland
  Lead AI Scientist at AMD Silo AI
  Email: susanna.pirttikangas@oulu.fi
  Web Page: oulu.fi/en/researchers/susanna-pirttikangas
- **Lauri Lovén**
  Assistant Professor (Tenure Track); Head of Future Computing Group, University of Oulu, Oulu, Finland
  Email: lauri.loven@oulu.fi
  Web Page: oulu.fi/en/researchers/lauri-loven